EXPERIMENT-2

AIM: Convert the ER Diagram for the Indian E-commerce Platform into a Relational Schema. Write Complete MYSQL CREATE TABLE Statements with Primary key, Foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and ON DELETE SET NULL. Insert Sample Data and Demonstrate referential integrity violation

1) **Relational Schema:**

Mapping Notes:

- Name (composite): First_Name, Last_Name in CUSTOMER.
- Phone (multi-valued, CUSTOMER only) → CUSTOMER_PHONE.
- OrderItem is a weak entity Identified by (Order ID, ProductID).
- SELLS is an M:N relationship between PRODUCT and SELLER → junction table PRODUCT_SELLER.
- The ISA specialization of PRODUCT (ELECTRONICS/CLOTHING/GROCERY) is mapped using one table per subtype, PK=FK= Product ID.

Table	Primary Key	Important Foreign Key
CUSTOMER	Customer_ID	—
CUSTOMER_PHONE	(CustomerID, Phone)	CustomerID → Customer
ADDRESS	AddressID	CustomerID → CUSTOMER
SELLER	SellerID	—
CATEGORY	Category ID	—
PRODUCT	ProductID	CategoryID → CATEGORY
PRODUCT_SELLER	(ProductID, SellerID)	ProductID → PRODUCT, SellerID → SELLER
ELECTRONICS	productID	ProductID → PRODUCT
CLOTHING	ProductID	ProductID → PRODUCT
GROCERY	ProductID	ProductID → PRODUCT
ORDERS	OrderID	CustomerID → CUSTOMER
PAYMENT	paymentID	OrderID → ORDERS
DELIVERY	DeliveryID	OrderID → ORDERS

2. SQL Queries

2.1 Schema Creation

```
DROP DATABASE IF EXISTS ECOMMERCE_ER;
CREATE DATABASE ECOMMERCE_ER;
USE ECOMMERCE_ER;

-- 1. CUSTOMER (Name is a composite attribute → First_Name,Last_N
-=ame)
CREATE TABLE CUSTOMER(
    CustomerID      INT  PRIMARY KEY,
    First_Name      VARCHAR(30)  NOT NULL,
    Last_Name       VARCHAR(30)  NOT NULL,
    Email           VARCHAR(100) NOT NULL UNIQUE,
    Date_of_Birth   DATE NOT NULL

-- Phone is a multi-valued attribute of CUSTOMER→ separate table
CREATE TABLE CUSTOMER_PHONE(
     CUSTOMERID INT,
     Phone   VARCHAR(15),
     PRIMARY KEY(CustomerID, Phone),
     FOREIGN KEY (CustomerID) REFERENCES CUSTOMER(customerID)
          ON DELETE CASCADE
  );

-- 2. ADDRESS (CUSTOMER 1 -HAS-- N ADDRESS,ADDRESS has total
-- participation)
CREATE TABLE ADDRESS(
     Address ID     INT PRIMARY KEY,
     House_No       VARCHAR(10),
     Street         VARCHAR(50),
     City           VARCHAR(30) NOT NULL,
     State          VARCHAR(30) NOT NULL,
     PIN_Code       VARCHAR(10)  NOT NULL,
     Address_Type   VARCHAR(20)
     CustomerID     INT NOT NULL,
     FOREIGN KEY (CustomerID) REFERENCES CUSTOMER(CustomerID)
          ON DELETE CASCADE
 );
```

```
-- 3. SELLER
CREATE TABLE SELLER(
    SellerID  INT PRIMARY KEY,
    Seller_Name  VARCHAR(50) NOT NULL,
    Email   VARCHAR(100) NOT NULL UNIQUE,
    Phone   VARCHAR(15),
    GSTIN   VARCHAR(20) NOT NULL UNIQUE
);

-- 4. CATEGORY
CREATE TABLE CATEGORY(
    CategoryID  INT PRIMARY KEY,
    CategoryName  VARCHAR(50) NOT NULL UNIQUE,
    Description  VARCHAR(150)
);
-- 5. PRODUCT (BELONGS_TO → CategoryID, nullable so SET NULL can be
- shown)
CREATE TABLE PRODUCT(
    ProductID  INT PRIMARY KEY,
    Product_Name  VARCHAR(100) NOT NULL,
    Price   DECIMAL(10,2) NOT NULL,
    Description   VARCHAR(200),
    Stock_Quantity  INT NOT NULL DEFAULT 0,
    Brand   VARCHAR(50),
    CategoryID  INT,
    FOREIGN KEY(CategoryID) REFERENCES CATEGORY(CategoryID)
        ON DELETE SET NULL
);
-- SELLS is an M:N relationship between PRODUCT and SELLER → junction
-- table
CREATE TABLE PRODUCT_SELLER(
    ProductID  INT,
    SellerID  INT,
    PRIMARY KEY(ProductID, SellerID),
    FOREIGN KEY(ProductID) REFERENCES Product(ProductID)
        ON DELETE CASCADE,
    FOREIGN KEY(SellerID) REFERENCES SELLER(SellerID)
        ON DELETE CASCADE
);
```

```
--6 ISA Specialization of PRODUCT → ELECTRONICS / CLOTHING / GROCERY
CREATE TABLE ELECTRONICS(
        ProductID  INT PRIMARY KEY,
        Warranty_Period  VARCHAR(30),
        Model_Number  VARCHAR(50),
        FOREIGN KEY(ProductID) REFERENCES PRODUCT(ProductID)
            ON DELETE CASCADE
);
CREATE TABLE CLOTHING(
        ProductID  INT PRIMARY KEY,
        Size  VARCHAR(10),
        Color  VARCHAR(30),
        Material  VARCHAR(30),
        FOREIGN KEY(ProductID) REFERENCES PRODUCT(ProductID)
            ON DELETE CASCADE
    );
    CREATE TABLE GROCERY(
        ProductID  INT PRIMARY KEY,
        Expiry_Date  DATE,
        Weight  DECIMAL(8,2)
        FOREIGN KEY(ProductID) REFERENCES, PRODUCT(ProductID)
            ON DELETE CASCADE
    );

    CREATE TABLE ORDERS (
        OrderID  INT PRIMARY KEY,
        Order_Date  DATE NOT NULL,
        Order_Status  VARCHAR(20) NOT NULL,
        Total_Amount  DECIMAL(10,2) NOT NULL,
        CustomerID  INT NOT NULL,
        FOREIGN KEY(CustomerID) REFERENCES CUSTOMER(customerID)
            ON DELETE CASCADE.
    );
    CREATE TABLE PAYMENT(
        PaymentID  INT PRIMARY KEY,
        Payment_Date  DATE  NOT NULL,
        Amount  DECIMAL(10,2)  NOT NULL,
        Payment_Method (VARCHAR(30) NOT NULL,
        TransactionID  VARCHAR(30)  UNIQUE,
```

```
    FOREIGN KEY(OrderID) REFERENCES ORDERS(OrderID)
        ON DELETE CASCADE
);

-- 8.DELIVERY (DELIVERED_BY: ORDER 1--1 DELIVERY)
CREATE TABLE DELIVERY(
    DeliveryID INT PRIMARY KEY,
    Delivery_Date DATE,
    Delivery_Status VARCHAR(20) NOT NULL,
    TrackingID VARCHAR(30) UNIQUE,
    Delivery_Partner VARCHAR(50),
    OrderID INT NOT NULL UNIQUE,
    FOREIGN KEY(OrderID) REFERENCES ORDERS(OrderID)
        ON DELETE CASCADE
);
```

2.2 Sample Data (INSERT)

```
USE ECOMMERCE_ER;

INSERT INTO CUSTOMER VALUES
(1, 'Jeevesh', 'Kumar', 'email1@gmail.com', '2006-01-01');

INSERT INTO CUSTOMER_PHONE VALUES
(1, '9214869101'),
(1, '08240111123');

INSERT INTO ADDRESS VALUES
(113, 'A', 'XYZ City', 'Lucknow', 'Uttar Pradesh', 'XXXX', 'Home', 1);

INSERT INTO SELLER VALUES
(101, 'TechStore India', 'tech@gmail.com', '9123456789', 'GST1234567');

INSERT INTO CATEGORY VALUES
(201, 'Electronics', 'Electronic gadget and accessories');

INSERT INTO PRODUCT VALUES
(301, 'Gaming Laptop', 75000.00, '15.6 inch, RTX Graphic', 10, 'ASUS', 201);

INSERT INTO ELECTRONICS VALUES
(301, '2Year', 'ASUS TUF-F15');
```

```
INSERT INTO PRODUCT_SELLER VALUES
(301,101);

INSERT INTO PAYMENT VALUES
(601, '2026-08-14', 75000.00, 'UPI', 'Paid', 'TxN123', 501);

INSERT INTO DELIVERY VALUES
(701, NULL, 'Processing', 'TRK123', 'Delivery', 501);
```

2.3. Referential Integrity Violation

```
INSERT INTO ORDER VALUES
(502, '2026-08-14', 'Placed', 50000.00, 999);
```

2.4 ON DELETE SET NULL DEMONSTRATION

```
DELETE FROM CATEGORY WHERE categoryID = 201;
```

2.5 ON DELETE CASCADE DEMONSTRATION

```
DELETE FROM CUSTOMERS WHERE CustomerID = 1;
```

OUTPUT

- **Output-1** – SHOW TABLES

```
SHOW TABLES;
```

Tables_in_ECOMMERCE_ER
ADDRESS
CATEGORY
CLOTHING
CUSTOMER
CUSTOMERPHONE
DELIVERY
ELECTRONIC
GROCERY
ORDERS
PAYMENT
PRODUCT
PRODUCT_SELLER
SELLER

- **Output-2** – Sample Data

```
SELECT * FROM CUSTOMERS;
```

CustomerID	First_Name	Last_Name	Email	Date_of_Birth
1	Jeevesh	Kumar	email1@gmail.com	2006-01-10

```
SELECT * FROM PRODUCT;
```

ProductID	Product_Name	Price	Stock_Qty	Brand	CategoryID
301	Gaming Laptop	75000.00	10	ASUS	201

```
SELECT * FROM ORDERS;
```

OrderID	Order_Date	Order_Status	Total_Amount	CustomerID
501	2026-08-14	Placed	75000.00	1

- **Output-3** — Referential Integrity Violation

ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint failed (`ECOMMERCE_ER`.`ORDERS`, CONSTRAINT `ORDER_ibfk_1` FOREIGN KEY (`customerID`) REFERENCES `CUSTOMER` (`customerID`) ON DELETE CASCADE)

- **OUTPUT-4** — ON DELETE SET NULL (CATEGORY - PRODUCT)

Before deletion:

ProductID	ProductName	CategoryID
301	Gaming_Laptop	201

After Deletion:

ProductID	ProductName	CategoryID
301	Gaming_Laptop	NULL

- **output-5** - ON DELETE CASCADE (CUSTOMER → everything Downstream)

Table	Row count after DELETE
ORDERS	0
ORDERITEM	0
PAYMENT	0
DELIVERY	0
ADDRESS	0
CUSTOMER_PHONE	0